

GP
 $\downarrow$
 $O(N)$ $\left[\begin{array}{l} \text{for (int } i=N; i>0; i=i/2) \\ \quad \text{for (int } j=0; j<=i; j++) \\ \quad \quad \text{print()}\end{array} \right]$ $\text{TC: } \text{Log } N$
 $\left[\begin{array}{l} \text{for (int } i=N; i>0; i=i/2) \\ \quad \text{print()}\end{array} \right]$
 $(\log N)$

$i=N$ n iterations
 $N/2$ $n/2$ iterations
 $N/4$ $n/4$ iterations

$i=0$ $\left(\frac{N}{2^k} \rightarrow 0 \right)$
 $N/2^{k-1} \approx 0$ $N < 2^k - 1$
 $\log N$
 Total iteration: $(n + n/2 + n/4 + n/8 + \dots)$
 $= n \left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{N} \right)$
 $1 + 1 \rightarrow 1 + 1$
 GP sequence
 $a=1$
 $r=1/2$
 $\text{iteration } x = \log n$

$$= n \left(\frac{a}{1-r} \right) \rightarrow 2n$$

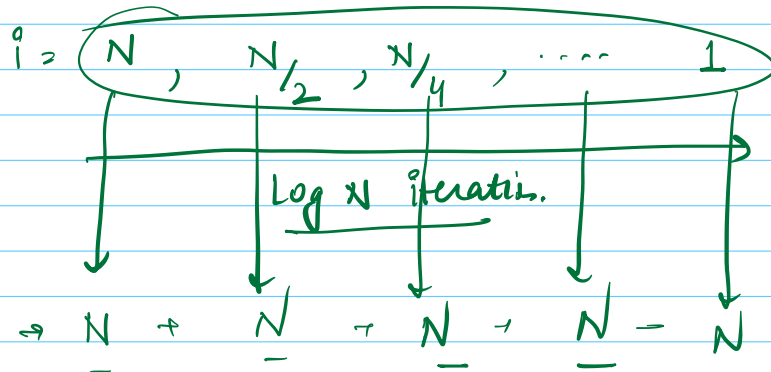
$$= \leq 2n \Rightarrow O(n)$$

$\log_2 N \Rightarrow$ It represents the no of times you would have to divide N by 2 so it reaches 1

Independent Loops

```
for ( int i=N ; i > 0 ; i=i/2 )
    for ( int j=0 ; j < N ; j++ )
        print()
```

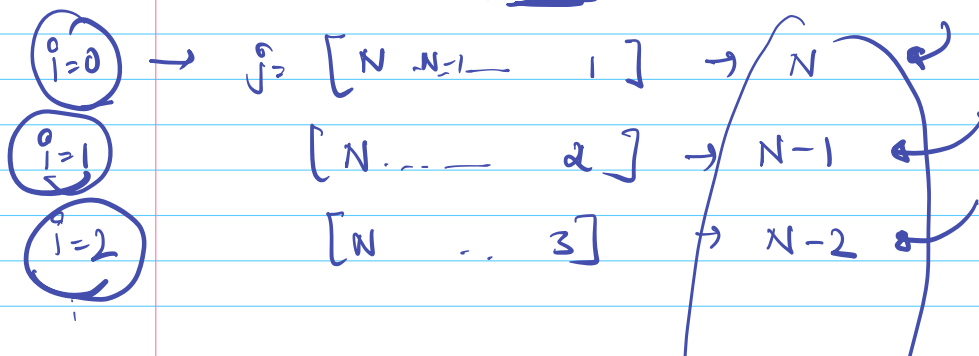
$O(n \log n)$

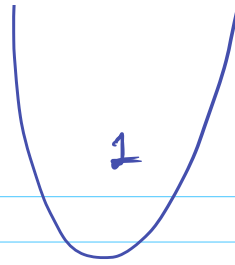


$N \log N$

```
for ( i=0 ; i < N ; i++ )
```

```
    for ( j=N ; j > i ; j-- )
        print()
```





$$1 + 2 + \dots + N = \frac{N * (N+1)}{2}$$

```

int count = 0
→ for (i = N; i > 0; i /= 2) ← log N
    for (j = 0; j < i; j++) ←
        count++

```

i = N j = [0, 1, 2, ..., N-1] → N times.

i = N/2 j = [0, 1, ..., $\frac{N}{2} - 1$] → $\frac{N}{2}$ times

i = N/4 → $\frac{N}{4}$ times

i = 1

Total iterations = $N + \frac{N}{2} + \frac{N}{4} + \dots + 1$

$$N + \frac{N}{2} + \frac{N}{4} + \dots + 1$$

$$N \left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{N} \right) \Rightarrow < 2N$$

$$\underbrace{\hspace{10em}}_{< 2}$$

$$\downarrow$$

$$\underline{\underline{O(N)}}$$

$$\textcircled{2} = \left(\frac{1}{1} + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots + \infty \right)$$

$$a = 1$$

$$r = \frac{1}{2}$$

$$S = \frac{a(r^n - 1)}{r - 1}$$

$$\begin{aligned} n &\rightarrow \infty \\ r &< 1 \end{aligned}$$

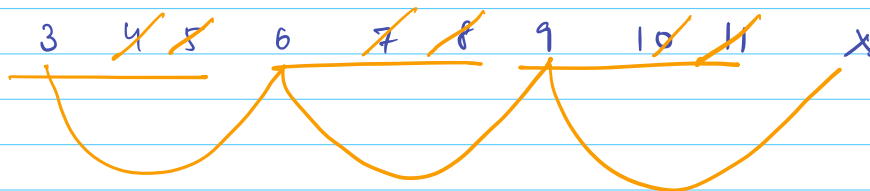
$$S = \frac{a}{1-r} = \frac{1}{1-\frac{1}{2}} = \textcircled{2}$$

Q1, 3, 7, 8, ~~25~~

for ($i=3$ $i < N/3$ $i+=3$)

$\sim N/3$ times

$i \neq 1$



$$(N/3)/3 = \textcircled{N/9}$$

for ($i=2$ $i < N/2$ $i+=2$) $\sim N/4$

$$N/2 \times N/4 = \underline{N^2} \Rightarrow O(N^2)$$

14 14 26 1 1

for (i = 0 ; i < N ; i++) $\Rightarrow$ N

for (i = 3 ; i < N ; i++) $\Rightarrow$ N - 3

for (i = 0 ; i < $N/3$; i++) $\Rightarrow$ $N/3$

for (i = 3 ; i < $N/3$; i++) $\Rightarrow$ $N/3$ - 3

$\rightarrow$ for (i = 0 ; i < N ; i += 3) $\Rightarrow$ $N/3$

for (i = 0 ; i < $N/3$; i += 3)

3

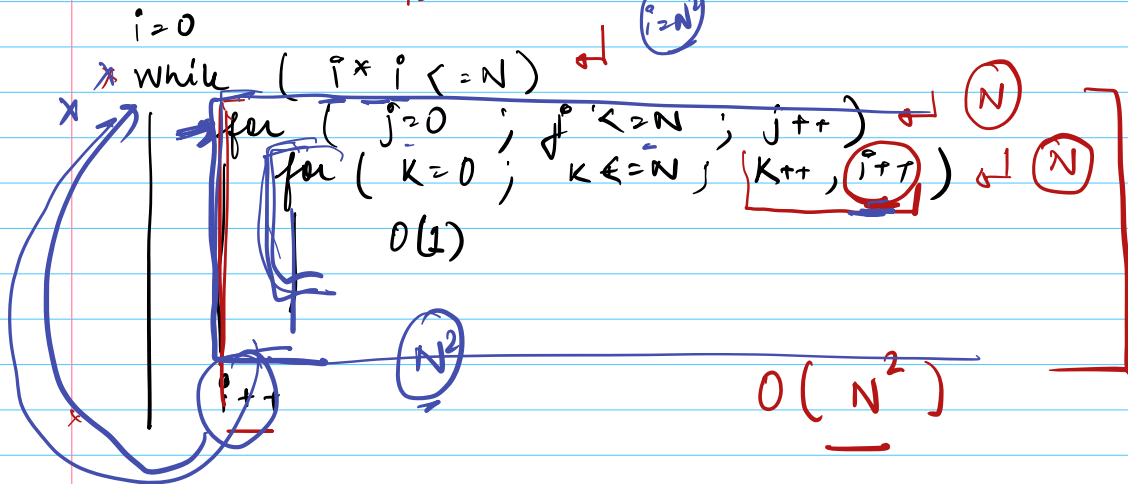
$N/9$

$N/4 - 3$ $N/9$

~~i = 0~~ $i \leq N$

~~i = 0~~

$i = N^2$



$i = 0$

$j = 0$

$k = 0 \rightarrow N$

N + 1 times

$j = 1$

$k = 0 \rightarrow N$

$i = N + 1$

$i = 2(N + 1)$

$j = 2$

$k = 0 \rightarrow N$

$i = 3(N + 1)$

,

$$j = N$$

$$K = 0 \rightarrow N$$

$$i = (N+1)(N+1)$$



```
for ( i = 0; i < 2^N; i++)
    while ( j > 0 )
        j = j - 1
```

$O(i)$

$j = 1 \rightarrow i \text{ times}$

$$[0 + 1 + 2 + 3 + \dots + 2^N]$$

$$\sum i$$

sum of 2^N natural n.s

$$\frac{2^N \times (2^N + 1)}{2}$$

$$2^{2N}$$

$$4^N$$

$$2^N$$

$$4^N$$

$$= 4^N$$

$$2^{10}$$

$$1024$$

$$4^{10}$$

$$10^6$$

$\sqrt{N}$

for ($i=1$; $i*i \leq n$; $i++$)

$O(1)$ for ($j=1$; $j*j \leq i$; $j+=i$)

single iteration $j=1$

$j=1$ $j=(1+i)$
 $(1+i) * (1+i) \leq i$

$j*j \leq i$

$j+=i$

$i*i \leq i$

Q2

$i=1$
 while ($i \leq N$)
 {
 $x=i$
 while ($x--$)
 $O(1)$
 $i++$
 } $\rightarrow i \text{ times}$

$i=1$
 1 times
 $i=2$
 +
 2 times
 $i=3$
 +
 3 times
 +

Sum of $N-1$ natural nos.

$i = N-1$
 $\rightarrow$ $N-i$ times

Q5

for ($i=0$; $i < N$; $i++$)
 $\rightarrow$ for ($j=i$; $j < N$; $j++$)
break
x
(N)

$i=0$

$j=0$

$i=1$

$j=1$

$i=2$

$j=2$

$i=3$

$j=3$